

Phishing Email Detection & Awareness Report

Internship:

Future Interns – Cber Security Task 2

Submitted By:

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Introduction

Phishing is one of the most common cyber attacks used by attackers to steal sensitive information from users. In phishing attacks, cybercriminals send fake emails that appear to come from trusted organizations, companies, or services. These emails are designed to trick users into clicking malicious links, downloading infected attachments, or sharing confidential information such as passwords, OTPs, and banking details.

Phishing attacks mainly rely on social engineering techniques like urgency, fear, fake notifications, and suspicious attachments to manipulate users. Many successful phishing attacks occur because users are unable to identify suspicious emails correctly.

This project focuses on analyzing phishing email samples and identifying common phishing indicators such as fake sender addresses, suspicious links, authentication failures, and malicious attachments. A legitimate email sample was also analyzed to compare the differences between safe and phishing emails.

The purpose of this project is to improve cybersecurity awareness and help users understand how phishing attacks work and how they can protect themselves from such threats. Through proper email analysis and awareness practices, users and organizations can reduce the risk of phishing attacks and data breaches.

Objectives

- Analyze phishing email samples
- Identify phishing indicators
- Classify email risk levels
- Compare phishing and legitimate emails
- Create cybersecurity awareness guidelines

Tools Used

- GitHub
- Google Message Header Analyzer
- MS Word
- Public phishing email repositories

Phishing Sample 1 Analysis :

1. Suspicious sender email

- The email is sent from attack@attacker.example.com.
- This is not an official MetroFax domain.

2. Fake fax notification

- The email pretends to be a SharePoint/MetroFax notification.
- Attackers commonly use fake fax or voicemail alerts.

3. Suspicious external link

- The "Print & Preview Here" link redirects to:
<https://attacker.example.com/>
- This is a malicious-looking domain.

4. Social engineering technique

- The message creates curiosity and urgency:
"You have a new fax!"

5. Unknown recipients

- "Undisclosed recipients" is suspicious and commonly used in phishing campaigns.

6. Possible malware delivery

- The email asks the user to open an attachment/PDF.
- Attachments in phishing emails may contain malware.

7. Impersonation attempt

- The email copies MetroFax branding and formatting to appear legitimate.

8. Suspicious formatting and hidden links

- HTML email with embedded hyperlinks can hide malicious destinations.

Risk Classification: PHISHING

Conclusion:

This email is a phishing attempt designed to trick users into clicking malicious links or opening infected attachments. Users should avoid interacting with the links or downloading files from this email.

Phishing Sample 2 Analysis :

1. SPF authentication failed

- The sender domain failed verification.
- This indicates the email may be spoofed or fake.

2. DKIM not configured

- The message was not digitally signed.
- This reduces trustworthiness of the email.

3. DMARC validation failed

- The email failed domain authentication checks.
- Strong indication of phishing activity.

4. Suspicious subject line

- "Voice Message-Access for Clients.Pass-Key-Exception"
- Designed to create urgency/confusion.

5. Unknown sender and suspicious voicemail notification

- Attempts to trick user into opening malicious content.

6. Suspicious instruction

- "Download attached file to listen to your voice message"
- Common malware/phishing delivery method.

7. Obfuscated text detected

- Characters like "vοice" are used to bypass spam filters.

8. Suspicious external image URL

- Email loads remote content from external source.

Risk Classification: PHISHING

Conclusion:

This email contains multiple phishing indicators including authentication failures, suspicious attachments, spoofing attempts, and social engineering tactics. Users should avoid downloading attachments or interacting with the message.

Legitimate Email Analysis :

1. Professional and clear language used
2. No suspicious links or attachments
3. No urgency or fear tactics detected
4. Sender purpose is clearly explained
5. No request for passwords or sensitive information

Risk Classification: SAFE

Conclusion:

This email appears legitimate because it uses professional communication, does not pressure the user, and contains no suspicious links or phishing indicators.

Phishing Prevention Guidelines

- Verify sender email addresses carefully
- Avoid clicking suspicious links
- Never share passwords or OTPs through email
- Do not download unknown attachments
- Check for spelling mistakes and urgency tactics
- Use multi-factor authentication whenever possible
- Report suspicious emails to the security team

Conclusion :

This project demonstrated how phishing emails use social engineering, fake domains, malicious links, and urgency tactics to deceive users. By analyzing phishing samples and comparing them with legitimate emails, important phishing indicators were identified successfully.

Cybersecurity awareness plays a major role in preventing phishing attacks. Users should remain cautious while handling emails, attachments, and links to reduce the risk of data theft and malware infections.